Good-equation two-soliton scattering at one loop: fixed-momentum counterterm response, three-soliton tests, and the unresolved antisymmetric phase

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4 November 2026

Abstract

(c) We independently evaluate the audited continuum integrals in both channels, verify three-soliton phonons by four-wave Hirota coefficient tests, and compute the response of the classical canonical scattering phase to $\Delta H = \lambda\hbar \int u^2 \mathrm{d}x$ at fixed physical momenta:

$$K_\lambda = 3\delta_0 + 4\sum_{j=1}^{2} P_j \Delta X_j' = \frac{2}{3}(v_2 - v_1)\log A_{12} + \text{finite endpoint terms.}$$

It is antisymmetric under exchange of the complete particle labels and is generically nonzero. Thus the canonical antisymmetric response is not scheme independent. We also give responses to finite changes of the kinetic, cubic and bending coefficients. A separate connected local-insertion calculation checks the logarithmic term.

(c) Scalar three-body factorisation and the exact factorised phonon transmission do not determine an unknown antisymmetric pair contribution. We exhibit a rank-two pair function that passes these tests; this is a counterexample to sufficiency of the tested constraints, not a freely adjustable physical coupling. **(d)** The boundary-compatible projected two- and three-soliton determinants and quantum state map remain uncomputed. Consequently this is a partial result, not the requested absolute physical one-loop phase. The JSON grid has 270 rows with explicit unknown physical entries, numerical ingredients, counterterm coefficients, and separate near-resonance data. No unknown is replaced by zero.

1 Scope and controlling evidence

Labels (a), (b), (c), (d) mean literature proof, literature heuristic, project derivation, and guess or unestablished identification. All new derived formulas below are checked by the script named in their section; scripts and caches are in `phase2/scripts/`. Exact algebra and finite numerical tests are distinguished. No legacy script was run or imported; only `good_twobody*` files were written. No renormalisation condition or quantum boundary prescription is selected.

(c) The controlling inputs are `audit_det_projected.tex`, `audit_projected_measure.tex`, `audit_determinant_foundations.tex`, `audit_smx.tex`, `audit_frame_dictionary.tex`, and `errata.tex`; the corresponding package claims are used only with their audit qualifications. In particular, neither the raw phonon integral nor its antisymmetrisation has been established as the full projected determinant. The Toda Krein equation is an additional microscopic input in a different model. It is not a consequence of scalar factorisation and is not imported into the good equation. This is a statement of the supplied audit record, not a new literature theorem.

The fixed convention is

$$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}, \qquad u = \phi_x = 2(\log\tau)_{xx},$$
$$L = \tfrac{1}{2}\phi_t^2 - \tfrac{1}{2}u^2 + u^3 - \tfrac{1}{2}u_x^2, \qquad \Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\phi, \quad \pi = \phi_t.$$

All real quantum candidates in the numerical grid are restricted to $0 < \alpha_j < \pi/3$, $A_{12} > 0$, and unequal velocities. Angles are in radians. A displayed ordered phase refers to particle 1 overtaking particle 2; exchanged labels denote the inverse oriented generator. Both $++$, $--$, $+-$, and $-+$ are saved.

2 Classical normalization and fixed-momentum conversion

Checked input (c) (`good_twobody_counterterms.py`, `good_twobody_kernels.py`). With $k_j = \sin\alpha_j$, $v_j = \epsilon_j \cos\alpha_j$,

$$M_j = 2k_j, \quad \Pi_j = -2k_j v_j, \quad P_j = \frac{2}{3} v_j k_j^3, \quad E_j = \frac{2}{3}k_j^3 - \frac{8}{15}k_j^5, \quad \ell_j = -\epsilon_j \cos 3\alpha_j.$$

The energy is the errata-corrected one. Define

$$p_{ja} = \tfrac{1}{2}(k_j + v_j/\sqrt{3}), \qquad p_{jb} = \tfrac{1}{2}(v_j/\sqrt{3} - k_j), \tag{1}$$

$$A_{12} = \frac{(p_{1a} - p_{2a})(p_{1b} - p_{2b})}{(p_{1a} - p_{2b})(p_{1b} - p_{2a})}, \qquad T_{12} = \frac{2}{3} M_1 M_2 (v_1 - v_2), \tag{2}$$

$$\delta_0 = \frac{2}{9}(\ell_1 - \ell_2)\log A_{12} + T_{12}, \qquad \gamma_j = -\frac{3\cos 2\alpha_j}{\sin\alpha_j \sin 3\alpha_j}, \tag{3}$$

$$\Delta X_1' = -\frac{\log A_{12}}{k_1} + \gamma_1 M_2, \qquad \Delta X_2' = \frac{\log A_{12}}{k_2} - \gamma_2 M_1. \tag{4}$$

Direct differentiation of the root-cross-ratio expression verifies $\partial_{P_j}\delta_0 = -\Delta X_j'$, without defining one side from the other. It also verifies antisymmetry of the derived responses under simultaneous exchange of angles and species.

Conditional interface (c) /(d) (`good_twobody_counterterms.py`, `good_twobody_grid.py`). First-order inversion of $P_j^{\mathrm{phys}} = P_j(\alpha_j) + \hbar p_j(\alpha_j) + O(\hbar^2)$ gives the algebraic conversion

$$\delta_{1,P} = \delta_{1,\alpha} + R[p], \qquad R[p] = \sum_j \Delta X_j' p_j. \tag{5}$$

The physical identification of both p_j and $\delta_{1,\alpha}$ is still **(d)** . The scripts retain two audited trace representatives to expose this issue:

$$p_{\mathrm{m}}(\alpha, \epsilon) = -\frac{\epsilon}{2\pi} \sum_r c_r \sin\varphi_r \log|\tan(\varphi_r/2)|, \tag{6}$$

$$e_{\mathrm{m}}(\alpha) = \frac{-6\sin\alpha - 2\sum_r c_r \sin\varphi_r \cos\varphi_r \log|\tan(\varphi_r/2)|}{4\pi}, \tag{7}$$

$$L_1 = v p_{\mathrm{m}} - e_{\mathrm{m}}, \qquad p_{\mathrm{can}} = \frac{\mathrm{d}L_1}{\mathrm{d}v}, \tag{8}$$

$$(\varphi_1, \varphi_2, \varphi_3) = (\alpha + 2\pi/3, \alpha - 2\pi/3, \alpha), \quad (c_1, c_2, c_3) = (1, 1, -2). \tag{9}$$

(c) The mode momentum is checked by independent half-line integration. The canonical representative is the derivative of the same conditional trace, not a new physical momentum determination. Switching from p_{m} to p_{can} requires changing the label-frame phase by $-R[p_{\mathrm{can}} - p_{\mathrm{m}}]$; otherwise it changes the answer. Agreement of the fixed-momentum one-body energy does not justify that inconsistent switch in a two-body phase.

3 Elastic and other bulk counterterm responses

Exact rescaling (c) (`good_twobody_counterterms.py`). Consider the family, with positive a, b, d near one,

$$H_{a,b,d,g} = \int \left(\frac{a}{2}\pi^2 + \frac{b}{2}u^2 - gu^3 + \frac{d}{2}u_x^2 \right) \mathrm{d}x. \tag{10}$$

The transformation

$$x = L_x X, \quad t = L_t T, \quad \phi = F\Phi, \qquad L_x = \sqrt{d/b}, \quad L_t = \frac{\sqrt{d/a}}{b}, \quad F = \frac{\sqrt{bd}}{g} \tag{11}$$

maps its action to s times the fixed action and its momentum to p_* times the fixed momentum, where

$$s = \frac{db^{3/2}}{\sqrt{a}\, g^2}, \qquad p_* = \frac{b^2\sqrt{d}}{\sqrt{a}\, g^2}, \qquad \delta_{a,b,d,g}(P_1, P_2) = s\, \delta_0(P_1/p_*, P_2/p_*). \tag{12}$$

The last expression uses the continuously rescaled classical Darboux coordinates and fixed laboratory momentum units. Every transformed Lagrangian coefficient and the independently transformed canonical momentum are checked symbolically. The crucial momentum factor for an elastic change is b^2, whereas the action factor is $b^{3/2}$.

(c) Set

$$a = 1 + 2\kappa\hbar, \quad b = 1 + 2\lambda\hbar, \quad g = 1 - \eta\hbar, \quad d = 1 + 2\zeta\hbar.$$

Thus $\Delta H/\hbar = \int(\lambda u^2 + \eta u^3 + \zeta u_x^2 + \kappa\pi^2)\mathrm{d}x$. With $D = \sum_j P_j \Delta X_j'$, expansion of the exact rescaling gives

$$K_\lambda = 3\delta_0 + 4D, \qquad K_\eta = 2\delta_0 + 2D, \tag{13}$$

$$K_\zeta = 2\delta_0 + D, \qquad K_\kappa = -\delta_0 - D. \tag{14}$$

All four are checked by numerically solving the exact deformed momentum equations before taking a central difference. At 75 digits and step 10^{-18}, the largest discrepancy is 1.87×10^{-35}, below the unchanged 10^{-32} assertion. This is a finite-difference validation error, not an uncertainty in an unknown quantum determinant.

(c) In particular,

$$\boxed{K_\lambda = \frac{2}{3}(v_2 - v_1)\log A_{12} + 3T_{12} + 4(P_1\gamma_1 M_2 - P_2\gamma_2 M_1).} \tag{15}$$

The response changes sign under complete label exchange; hence its symmetric part vanishes. At $(\alpha_1, \alpha_2) = (0.3, 0.6)$,

$$K_\lambda^{++} = 0.05008140486841890958\ldots, \qquad K_\lambda^{+-} = -0.22517393868334175848\ldots.$$

Consequently there is no general λ-independence of the antisymmetric canonical phase. The statement is for the specified classical-frame response, with other bulk and boundary couplings held fixed. It is not a claim that an uncomputed quantum boundary prescription is independent of all possible scheme changes.

Independent connected insertion (c) (`good_twobody_local_insertion.py`). The bilinear equation implies, for $f = \log\tau$, $f_{tt} - f_{xx} + f_{xxxx} + 6f_{xx}^2 = 0$. Let $F(t)$ be the spatial integral of $\log\tau_{12} - \log\tau_1 - \log\tau_2$, with the time-independent right plateau $\log A_{12}$ subtracted. Then

$$\int \mathrm{d}t\mathrm{d}x\,[u_{12}^2 - u_1^2 - u_2^2] = -\frac{2}{3}[F'(+\infty) - F'(-\infty)] = \frac{2}{3}(v_1 - v_2)\log A_{12}. \tag{16}$$

Indeed $F'(-\infty) = -v_2\log A_{12}$ and $F'(+\infty) = -v_1\log A_{12}$. Independent spatial quadratures at $t = \pm 3000$ test these slopes in both channels. The minus sign from $\Delta L = -\lambda\hbar u^2$ reproduces the logarithmic part of (15). The remaining finite part is the difference between the connected bulk insertion and the response of the canonical scattering generator, including its changing endpoint/Darboux terms. Dropping those terms would not implement the stipulated fixed canonical momenta and frame. Their quantum state interpretation is not resolved here.

What "other allowed counterterms" means. (c) (`good_twobody_counterterms.py`, `good_twobody_threebody.py`) A finite tadpole $\mu\hbar\int u\mathrm{d}x$ has zero connected scattering insertion: the exact tau asymptotes give $M_N=\sum_j 2k_j$ at every time. Vacuum constants also cancel after vacuum normalization. For a uniform background U, the large-momentum expansion of the one-loop energy integrand is checked directly from

$$\tfrac{1}{2}\sqrt{q^2+q^4-6Uq^2}=\tfrac{1}{2}|q|\sqrt{1+q^2}-\frac{3U|q|}{2\sqrt{1+q^2}}-\frac{9U^2|q|}{4(1+q^2)^{3/2}}+O(U^3). \tag{17}$$

Only its term linear in U is UV divergent in one spatial dimension. For a one-loop background ring with n insertions the superficial anisotropic degree is $3-2n$: each insertion contributes at most two internal spatial derivatives, each propagator degree -4, and the frequency/momentum measure degree 3. Thus no further bulk divergent counterterm is forced at this order in the specified cubic theory. These are perturbative UV statements, not infrared or boundary renormalisation results.

(d) Symmetries alone do not restrict finite local modifications to (14). Terms such as $\hbar c_n u^n$ and higher derivative terms are possible changes of the theory; boundary operators are a separate issue. Their finite coefficients are not fixed by the given data. We parameterise their possible contribution as $\sum_A c_A K_A$, with K_A *uncomputed* outside the four-coupling family above, rather than assert an exhaustive finite classification. No new higher interaction is silently set to a preferred value. The response K_λ is ready to accept the parallel renormalisation package's coefficient without choosing that coefficient.

4 Continuum integrals and the symmetric/antisymmetric split

Definitions and independent checks (c) (`good_twobody_kernels.py`, `good_twobody_grid.py`)**.** Use the Jost phase continuous on each open half-line and vanishing at infinity. With the roots and weights of (9),

$$\theta(q;\alpha,\epsilon)=-\sum_r c_r\Bigg[\operatorname{sgn}(q)\arctan\frac{\sin\varphi_r}{|q|} + \epsilon\arctan\frac{\sin\varphi_r\cos\varphi_r}{|q|\sqrt{1+q^2+q^2+\sin^2\varphi_r}}\Bigg], \tag{18}$$

$$\theta'(q)=\sum_r\frac{c_r\sin\varphi_r}{q^2+\sin^2\varphi_r}\left(1+\frac{\epsilon\operatorname{sgn}(q)\cos\varphi_r}{\sqrt{1+q^2}}\right). \tag{19}$$

The transmission obtained directly from the KP-root cross ratio agrees with $e^{i\theta}$. Differentiation of the arctangent formula agrees with the independently written derivative. Define ordinary half-line integrals, with no delta function inserted at zero,

$$I_{12}=\frac{1}{4\pi}\int_{\mathbb{R}\setminus\{0\}}\theta_1\theta_2'\mathrm{d}q,\qquad J_{12}=\frac{I_{12}-I_{21}}{2},\qquad I_{12}^{\mathrm{sym}}=-\frac{\pi}{4}(\epsilon_1+\epsilon_2). \tag{20}$$

The last equality follows from the endpoints $\theta(0^\pm)=\pi(\pm1+\epsilon)$. Independent quadrature for all four orientations gives maximum closed-form discrepancy 5.94×10^{-66} at 65 digits.

(c) For reproducibility the closed form uses

$$L(z)=\int_0^z\frac{2\log|t|}{1-t^2}\mathrm{d}t,\qquad L(1)=-\pi^2/4,\quad L(-z)=-L(z), \tag{21}$$

$$I^{++}(\alpha_1,\alpha_2)=\frac{1}{2\pi}\sum_{rs}c_rc_sL\left(\frac{\tan(\varphi_{1r}/2)}{\tan(\varphi_{2s}/2)}\right), \tag{22}$$

$$I^{+-}(\alpha_1,\alpha_2)=\frac{1}{2\pi}\sum_{rs}c_rc_sL\left(\tan(\varphi_{1r}/2)\tan(\varphi_{2s}/2)\right), \tag{23}$$

$$I^{\epsilon_1\epsilon_2}=\tfrac{1}{2}(\epsilon_1+\epsilon_2)I^{++}+\tfrac{1}{2}(\epsilon_1-\epsilon_2)I^{+-}. \tag{24}$$

These are real half-line formulas, not a prescription for holomorphic crossing beyond their stated domain. In particular I and J are alternative representatives sharing a local gradient, not additive pieces of the phase. The raw symmetric term is an endpoint artifact; it is not a physical symmetric scattering phase established by this work.

Exactly what is independent of λ. **(c)** At one-loop order the displayed undeformed-background integrals I, J, their endpoint symmetric term, and the reference-trace conversion $R[p_{\mathrm{m}}]$ are λ-independent. An order-$\hbar$ deformation of their arguments affects the scattering phase only at the next loop order. All explicitly computed local λ dependence at this order is λK_λ in the rescaled classical frame. The antisymmetric J alone is independent of λ; the antisymmetric canonical response is not. Nothing here proves scheme independence of the unknown boundary contribution under more general changes of the theory.

(d) A useful but conditional parametrisation is

$$\delta_{1,P} = J_{12} + R[p_{\mathrm{m}}] + \lambda K_\lambda + \eta K_\eta + \zeta K_\zeta + \kappa K_\kappa + B_{12} + \sum_A c_A K_A. \tag{25}$$

Here B_{12} includes the missing boundary/collective determinant, state map, and absolute normalization relative to the displayed trace representative. It is not known to be a constant. If scalar exchange is identified with inverse physical scattering, a locally continuous unitary representative has symmetric part zero, and $B_{21} = -B_{12}$ modulo the phase branch. This algebraic parametrisation does not prove that the quantum theory is elastic or factorised.

Channel	(α_1, α_2)	J	$R[p_{\mathrm{m}}]$	$R[p_{\mathrm{can}}]$	K_λ
$++$	$(.2, .4)$	.412598425	2.479485715	3.699829027	.064969423
$++$	$(.3, .6)$	.392706352	1.968010544	3.666344128	.050081405
$++$	$(.4, .6)$	.227909672	.811937649	2.079893527	.057940143
$+-$	$(.2, .4)$	.217194901	4.825323137	6.747182050	.061623021
$+-$	$(.3, .6)$	.408542241	4.915210065	8.209382585	-.225173939
$+-$	$(.4, .6)$	.478642156	5.241126358	9.896087545	-1.073366855

(c) (`good_twobody_grid.py`) These are ingredients, not two competing physical answers obtained by selecting a preferred momentum convention. For example $J + R[p_{\mathrm{m}}] = 2.360716896635\ldots$ at $(.3, .6, ++)$; its difference from the full phase has no numerical error bound because B_{12} is uncomputed.

5 Three-soliton test: what factorisation does and does not fix

Exact mode construction (c) (`good_twobody_threebody.py`). For $E_j = \exp[k_j(x - v_j t - x_{0j})]$, define

$$\tau_N(E) = \sum_{\mu \in \{0,1\}^N} \prod_j E_j^{\mu_j} \prod_{i<j} A_{ij}^{\mu_i \mu_j}. \tag{26}$$

Adding a fourth infinitesimal wave $K = iq$, $W = i|q|\sqrt{1+q^2}$ to $N = 3$ gives

$$\delta\phi = 2\partial_x \left[e^{iqx - i\omega t} \frac{\tau_3(T_1 E_1, T_2 E_2, T_3 E_3)}{\tau_3(E_1, E_2, E_3)} \right], \qquad T_j = A(j, \mathrm{ph}), \quad \omega = |q|\sqrt{1+q^2}. \tag{27}$$

The coefficient test expands the full four-wave bilinear equation using $\mathcal{P}(K, W) = W^2 - K^2 + K^4$ and $A_{ij} = -\mathcal{P}(K_i - K_j, W_i - W_j)/\mathcal{P}(K_i + K_j, W_i + W_j)$. It checks all 81 exponential coefficients exactly for a rational three-soliton configuration and one imaginary phonon. Independent 75-digit checks cover three right-movers and a mixed-direction triple, with both signs of phonon momentum. This is explicit coefficient evidence, not a spectral completeness proof.

(c) Exact polynomial clustering gives

$$\tau_3|_{E_3=0} = \tau_2(E_1, E_2), \tag{28}$$

$$\lim_{E_3\to\infty} \frac{\tau_3}{E_3} = \tau_2(A_{13}E_1, A_{23}E_2), \tag{29}$$

$$\lim_{E_3\to\infty} \frac{\tau_3(T_jE_j)}{\tau_3(E_j)} = T_3\frac{\tau_2(T_1A_{13}E_1, T_2A_{23}E_2)}{\tau_2(A_{13}E_1, A_{23}E_2)}. \tag{30}$$

Thus the continuum transmission is $\prod_j T_j$. In a sequence of separated binary collisions, the *formal continuum representative* is the sum of the oriented J_{ij}. This is what the script records; the full three-body determinant entry is `null`.

A precise insufficiency result (c) (`good_twobody_threebody.py`). Suppose a scalar three-body amplitude has the stipulated cluster form $S_3 = S_{12}S_{13}S_{23}$. For any real antisymmetric pair function F_{ij}, the replacements

$$S_{ij} \longmapsto e^{iF_{ij}}S_{ij}, \qquad S_3 \longmapsto e^{i(F_{12}+F_{13}+F_{23})}S_3 \tag{31}$$

preserve that factorisation and scalar collision-order consistency. An explicit example is

$$F_{12} = c(P_1M_2 - P_2M_1), \qquad c \in \mathbb{R}. \tag{32}$$

It is nonzero, antisymmetric, vanishes on a zero-charge leg, and has rank at most two as a kernel of one-body charges. Its 3×3 minors vanish identically. It also changes sign under the formal map $(P, M) \mapsto (-P, -M)$ on either leg. This last algebraic property is not an asserted physical crossing axiom for the good theory.

The example does not evade the Toda audit: that audit additionally has a microscopic Fredholm/Krein equation, and its excluded example has infinite rank. Neither the finite-rank condition nor scalar clustering by itself eliminates (32). A fully determined Boussinesq determinant gradient would eliminate additions that change that gradient; the point is that such a gradient has not been established on the physical constrained domain.

(d) Thus computing the exact three-soliton *phonon modes* and imposing clustering does not compute its one-loop *phase*, and does not fix B_{12} or the residual ϑ. This result concerns sufficiency of the proposed test, not a proof of arbitrary physical freedom. Spatial removal of a spectator at finite observation time must also be distinguished from an infinite-time scattering experiment in which that spectator eventually collides.

The boundary obstruction persists (c) /(d) (`good_twobody_threebody.py`). For the centered potential $\Phi = k\tanh(kx/2)$,

$$\Phi_\alpha = \cos\alpha\,[\tanh(kx/2) + (kx/2)\operatorname{sech}^2(kx/2)], \qquad \|\Phi_\alpha\|^2_{[-R,R]} = 2R\cos^2\alpha + O(1). \tag{33}$$

The tail slope is rechecked by independent quadrature. The controlling projector audit gives localized data whose collective projection acquires a nonzero endpoint tail. Those collective and radiation directions cannot be integrated independently with both held in a fixed-endpoint domain. Even at finite dimension, $\det \left(\begin{smallmatrix} A & B \\ B^T & C \end{smallmatrix}\right) = \det C \det(A - BC^{-1}B^T)$; knowledge of the radiation block does not compute the Schur factor. The scripts check this identity in a nontrivial scalar-block example. **(d)** A common domain, propagated constraints, completeness, the Schur factor, and temporal state normalization remain necessary. No full determinant or relative Maslov index is claimed here.

6 Approach to resonance and cached data

Regular head-on Y-boundary (c) (`good_twobody_grid.py`). For $\epsilon_1 = +1, \epsilon_2 = -1$, fix $0 < \alpha_1 < \pi/3$ and put $\alpha_2 = \pi/3 - \alpha_1 - h$, $h \downarrow 0$. The root denominator vanishes and $A_{12} \to +\infty$. The integrals have a finite limit, checked by direct endpoint quadrature. For either specified momentum function p,

$$J + R[p] = C_p \log A_{12} + B_p + o(1), \tag{34}$$

$$C_p = [-p_1/k_1 + p_2/k_2]_{h=0}\,, \qquad B_p = [J + \gamma_1 M_2 p_1 - \gamma_2 M_1 p_2]_{h=0}\,, \tag{35}$$

$$K_\lambda = C_\lambda \log A_{12} + B_\lambda + o(1), \qquad C_\lambda = \left[\tfrac{2}{3}(v_2 - v_1)\right]_{h=0}\,, \tag{36}$$

$$B_\lambda = [3T_{12} + 4(P_1\gamma_1 M_2 - P_2\gamma_2 M_1)]_{h=0}\,. \tag{37}$$

At $\alpha_1 = 0.4$, the checked numbers are

$$J_* = 0.516497248493590315982850593057\ldots,$$
$$C_{p_\mathrm{m}} = 1.271224247490672273785345801 75\ldots,$$
$$C_\lambda = -1.145891778762162276063217330 46\ldots.$$

Therefore the logarithmic coefficient of the conditional mode representative is $C_{p_\mathrm{m}} + \lambda C_\lambda$ when the other computed bulk changes are absent. **(d)** The unknown boundary phase can change the full asymptotics; this is not a prediction of a physical resonant amplitude or fusion rule. The expansion is taken at fixed regular angles before the resonance limit and is not asserted uniform when $\hbar|\log h|$ becomes order one.

(c) (`good_twobody_grid.py`) The genuine overtaking Y-line is $\alpha_1 + \alpha_2 = 2\pi/3$. It has no interior point with both legs on the light branch. We save classical responses approaching this line with one heavy leg, with quantum phases explicitly absent. The light overtaking $A \to 0$ limit on the diagonal is instead an equal-velocity threshold; the sampled K_λ tends to zero, not to a nonzero logarithmic divergence. The historical $|\alpha_1 - \alpha_2| = \pi/3$ overtaking resonance is false: $A(0.2, 0.2+\pi/3) = 0.355465479849\ldots$, finite and nonzero. The grid does not analytically continue a real one-loop phase through an unstable heavy branch or a singular $A < 0$ tau function.

Files and error budget (c) (all five scripts). `good_twobody_grid.json` covers the Cartesian angles $0.1, 0.2, \ldots, 1.0$, excluding equal velocities and nonpositive head-on A, in all four orientations: 270 rows. Each row is recalculated at 55 and 75 digits; the largest difference in its tested numerical fields is 8.31×10^{-55}. The independent quadrature tolerance is 10^{-55}, the fixed-momentum finite-difference tolerance 10^{-32}, and tabulated decimals are rounded. These are deterministic numerical consistency estimates, not rigorous interval enclosures. They do not cover the unknown physical determinant, whose uncertainty is unbounded by this calculation.

`good_twobody_resonance.json` contains three head-on approaches at $\alpha_1 = 0.2, 0.4, 0.5$ with $h = 10^{-2}, 10^{-4}, \ldots, 10^{-12}$, limiting coefficients and subtracted remainders, plus the overtaking threshold and heavy Y-line diagnostics. At the last head-on point the maximum discrepancy from the subtracted limits is below 10^{-8}; individual finite-distance errors are stored rather than identified with quadrature errors. The other four JSON files hold kernel, scaling, mode, and local-insertion verification results. Strings retain 55 significant digits; `null` means uncomputed, never zero.

7 Claim/check summary and failures recorded

Claim and status	New checker
(c) Continuum I, J, all orientations; endpoint symmetric term	`good_twobody_kernels.py`; `good_twobody_grid.py`
(c) Fixed-momentum response to u^2, u^3, u_x^2, π^2; antisymmetry	`good_twobody_counterterms.py`
(c) Connected u^2 insertion and canonical endpoint difference	`good_twobody_local_insertion.py`
(c) Four-wave coefficient tests; exact three-soliton clustering	`good_twobody_threebody.py`
(c) Scalar clustering is insufficient to fix the pair phase	`good_twobody_threebody.py`, rank-two example
(c) Regular grid and qualified resonance asymptotics	`good_twobody_grid.py`
(d) Complete projected determinant, absolute physical phase, quantum state map	Not computed; explicit unknown entries
(d) Other finite local interactions and quantum resonant channels	Not determined by the supplied theory and checks

Failures recorded.

1. The first connected-insertion quadrature subtracted three nearly constant temporal log-tau derivatives numerically on an infinite interval. Its unchanged 10^{-35} slope test failed at the overtaking point, $t = -3000$, with error $8.2718016287 \times 10^{-4}$. The plateau cancellation was performed algebraically before integration, and equality of the original and stabilized integrands was proved with SymPy. The same test and tolerance were retained. This failure is not hidden by the final success markers.

2. An initial hand scaling assigned momentum the action factor $b^{3/2}$. Direct transformation of $P = -\int \pi \phi_x \mathrm{d}x$ corrected it to b^2 before the numerical checks. The exact transformed momentum and independent inverse-momentum tests are included to prevent recurrence.

3. The intended inference "factorised three-soliton phonons plus clustering fixes the quantum antisymmetric phase" fails as a logical completion strategy: (31)–(32) preserve the tested constraints. This is a negative result, not a failed numerical test repaired by deleting a constraint. The full three-body phase was not computed.

4. The inherited identification of a continuum integral with the absolute determinant, the old false overtaking resonance line, and the uncorrected classical energy are not adopted. Their controlling audits and errata retain precedence. No check was weakened and no legacy failure was claimed to have been repaired by this package.

5. The first PDF build had two overwide equation lines and one overwide command line. They were split; an orphan section heading and a split reading-scope paragraph were moved to fresh pages before the final two-pass build and rendered inspection. Scientific formulas were unchanged.

Verification commands. Run each of the five named scripts with the environment variable `PYTHONDONTWRITEBYTECODE=1` and command `python3 phase2/scripts/`*script*. Each successful run ends with `ALL CHECKS PASSED` only after its assertions. Total runtime is comfortably below 30 minutes on this machine; the 270-row grid took about 107 seconds. Build this note by running `pdflatex -interaction=nonstopmode -halt-on-error good_twobody.tex` twice in `phase2/`. No project-wide script loop is needed.

Reference verification and reading scope

The repository inputs listed in Section 1 were read, with particular attention to the audit verdicts. For the external reference below, the local PDF title page and relevant full-text passages on printed pages 52–53, 62 and 65–66 were read; the entire paper was *not* read for this package. No new arXiv identifier or DOI is asserted. **(a)** Faddeev–Korepin's discussion explicitly concerns a Klein–Gordon fluctuation determinant and leaves a factor Z to supplementary requirements in its two-body formula. **(d)** Applying its full determinant argument to the constrained fourth-order Boussinesq problem remains unproved here; the reference is not cited as that missing theorem.

References

[1] L. D. Faddeev and V. E. Korepin, *Quantum theory of solitons*, Physics Reports **42**, no. 1 (1978), 1–87. [Local PDF, title page and specified full-text passages read.]